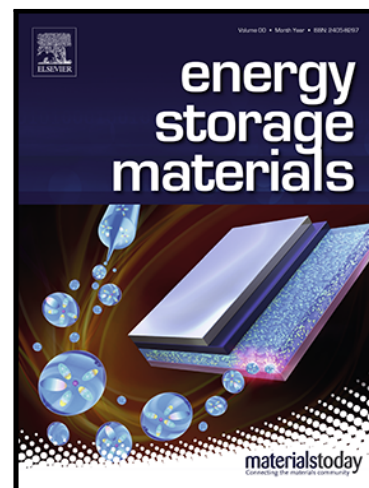


Bilayer Artificial Interface Engineering Enables Dendrite-Free and Low-Temperature Stable Lithium Metal Batteries

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PII: S2405-8297(25)00662-2
DOI: <https://doi.org/10.1016/j.ensm.2025.104664>
Reference: ENSM 104664



To appear in: *Energy Storage Materials*

Received date: 8 July 2025
Revised date: 28 September 2025
Accepted date: 6 October 2025

Please cite this article as: Yahui Li , Kai Yang , Zengjie Fan , Chong Xu , Zhemin Li , Hui Dou , Xiaogang Zhang , Bilayer Artificial Interface Engineering Enables Dendrite-Free and Low-Temperature Stable Lithium Metal Batteries, *Energy Storage Materials* (2025), doi: <https://doi.org/10.1016/j.ensm.2025.104664>

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Highlights

- . Bilayer ZnO/Li₂O coatings enable uniform and dendrite-free Li metal deposition.
- . Li₂O-rich interfaces induce selective TFSI⁻ reduction, forming LiF-rich SEI layers.
- . LiZn alloy acts as both Li reservoir and nucleation site for guided bottom-up growth.
- . The engineered interface ensures stable cycling performance even at low temperatures.

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Abstract

Rational design of lithium (Li) metal anodes is critical for achieving uniform Li deposition and long-term cycling stability. Here, we report a sequential atomic layer deposition of lithiophilic ZnO and Li₂O coatings on 3D carbon paper to construct a vertically graded bilayer artificial interface. The underlying ZnO-derived LiZn/Li₂O domains act as mixed ionic/electronic conductors and provide stable nucleation sites for Li deposition. The high-resistance Li₂O top layer suppresses electron tunneling and increases the nucleation size. This synergistic design directs bottom-up Li growth, fully exploiting the internal voids of the 3D scaffold, suppressing dendrite formation. As a result, the Li@CP/Z6L1||LCO cell achieves 83.1% capacity retention after 200 cycles at -20 °C (~4000 h), while the Li@CP/Z6L1||LFP cell maintains 96.4% capacity after 500 cycles, with only 0.0072% capacity loss per cycle. This work demonstrates a rational bilayer interface design that leverages vertical conductivity gradients and interfacial chemistry to achieve high-performance, durable, and low-temperature-tolerant lithium metal anodes.

Keywords: Lithium Metal Batteries, 3D current collector, Bilayer artificial interface, Dendrite-free anode, Low-temperature

1. Introduction

Lithium metal batteries (LMBs) have emerged as a transformative energy storage technology, leveraging the unparalleled theoretical capacity (3860 mAh g^{-1}) and ultra-low electrochemical potential (-3.04 V vs. SHE) of lithium metal anodes (LMA)[1-3]. However, the uncontrollable growth of Li dendrites and drastic volume fluctuations during cycling remain formidable barriers to practical applications. These issues lead to rapid capacity decay, electrolyte depletion, and catastrophic safety hazards. Achieving dendrite-free Li deposition is essential for ensuring the safe operation and long-term cycling stability of LMBs. The root cause of the unlimited volume expansion lies in the host-free planar deposition of Li metal[4, 5]. In LMB systems, planar electrodes tend to generate sharp protrusions during the initial stages of deposition, which locally intensify Li^+ flux. The uneven distribution of Li^+ flux directly triggers the formation of Li dendrites. To address this problem, the rational design of Li metal hosts that provide ample accommodation space and promote uniform Li^+ flux distribution has emerged as a promising strategy[6, 7]. Three-dimensional (3D) current collectors with high surface area not only reduce local current density to homogenize Li deposition but also mitigate volume expansion induced by Li metal. To date, commonly reported 3D current collectors, including metallic foams[8-10], carbon nanotubes[11-13], carbon fibers[14-16], and 3D MXene-based architectures[17-19], have been extensively explored to promote uniform Li deposition.

Carbon-based hosts are regarded as highly promising candidates for dendrite-free LMA due to their chemical stability, excellent electronic conductivity, and lightweight

properties[20-22]. For example, Cui et al. reported a layered Li-rGO electrode fabricated by infusing molten Li into the nanoscale gaps of lithiophilic reduced graphene oxide (rGO)[23]. The periodic stacking of nanosized Li and rGO layers provides a robust scaffold for Li plating and stripping, effectively mitigating volume fluctuations during cycling. Similar to this strategy, previous studies have typically employed molten Li infusion to prepare composite Li anodes, which inevitably leads to safety concerns, interfacial inhomogeneity, and imprecise control of the Li content [24-26]. In addition, most carbon materials exhibit relatively weak interactions with Li, resulting in high nucleation overpotentials[27, 28]. As a result, Li tends to deposit laterally along the top surface rather than penetrating vertically into the porous structure. This inefficient space utilization exacerbates dendrite formation and significantly limits the advantages of 3D frameworks.

Recent studies have demonstrated that rationally designing the conductivity and lithiophilicity gradients of 3D current collectors can effectively address the challenge of Li growth concentrated at the top surface[29, 30]. For instance, Wu et al. applied ZnO and Au coatings on the top and bottom surfaces of copper foam, respectively[8]. This hierarchical structure enabled a gradual enhancement of lithiophilicity and conductivity from top to bottom, providing sufficient driving force for the downward migration of Li^+ . Consequently, Li was synchronously deposited throughout the entire 3D space. In addition, Oyakhire and co-workers systematically investigated Cu current collectors modified with nanofilms of different resistivities to explore the correlation between substrate electrical properties and Li deposition behavior[31]. They found that

high-resistivity surfaces favored the formation of larger Li nuclei, which effectively reduced the specific surface area of the deposits, thereby mitigating electrolyte consumption and suppressing subsequent dead Li accumulation. However, rationally designing artificial coatings to induce bottom-up Li deposition within 3D hosts remains a significant challenge.

Herein, we propose a bilayer artificial interface engineering strategy by sequentially depositing ZnO and Li₂O layers via ALD on carbon paper (CP), denoted as CP/ZnO/Li₂O (The optimized sample is referred to as CP/Z6L1), and further fabricating composite Li anodes by cold-pressing with 50 μ m Li foil. The underlying ZnO-derived LiZn/Li₂O domains act as mixed ionic/electronic conductors, which provide low-energy nucleation sites and facilitate the rapid lateral electron transport, guiding the growth of Li metal along the Li (110) crystal plane parallel to the substrate[32]. Moreover, Li₂O functions as an electron-insulating yet ion-conductive mediator, effectively suppressing electron tunneling at the LiZn/Li₂O and Li₂O interface while enhancing Li⁺ transport[33, 34]. Density functional theory (DFT) calculations confirm that the terminal oxygen groups of Li₂O act as electron donors, exhibiting nucleophilic characteristics that promote the preferential reduction of TFSI⁻ anions to form a LiF-rich SEI layer[3]. The constructed vertical-gradient bilayer artificial interface guides bottom-up Li growth, fully exploiting the internal voids of the 3D scaffold and suppressing dendrite formation. This synergistic design enhances the utilization efficiency of the entire 3D framework for Li deposition, thereby alleviating volume fluctuations of the anode. As a result, the optimized CP/Z6L1@Li symmetric cell

demonstrates an ultralong cycling life of 2900 hours with an overpotential below 10 mV. Even at a high current density of 8 mA cm^{-2} , the overpotential remains as low as 40 mV, significantly lower than the 78 mV observed for CP. Additionally, when paired with a high-loading LiCoO_2 cathode, the full cell maintains a capacity retention of 83.1% after 200 cycles at -20°C , maintaining stable cycling for nearly 4000 hours. Moreover, the Li@CP/Z6L1||LFP full cell exhibited excellent cycling stability, retaining 96.4% of its capacity after 500 cycles at 1 C, corresponding to an average capacity loss of only 0.0072% per cycle.

2. Results and discussion

During the ALD process, diethylzinc and H_2O precursors were sequentially introduced onto the carbon paper (CP) surface to deposit a ZnO layer, followed by Li tert-butoxide treatment to form an overlying Li_2O coating (Fig. 1a). The synergistic lithiophilic nature of ZnO and Li_2O collectively facilitates uniform Li nucleation and deposition. To quantitatively assess lithiophilicity, density functional theory (DFT) calculations were performed to determine the adsorption energies of Li atoms on various surfaces. The calculated adsorption energies for Li (110), ZnO (101), and Li_2O (111) surfaces were -3.731 , -5.364 , and -4.023 eV, respectively, in comparison to -1.400 eV on pristine carbon paper (Fig. 1b). The more negative adsorption energies on ZnO and Li_2O indicate significantly enhanced Li affinity, suggesting preferential Li^+ nucleation on these modified surfaces. Furthermore, differential charge density analysis revealed substantial electron accumulation at the Li/ZnO and Li/ Li_2O interfaces (Fig. S1), indicating strong interfacial electronic interactions that contribute to the

stabilization of Li deposition.

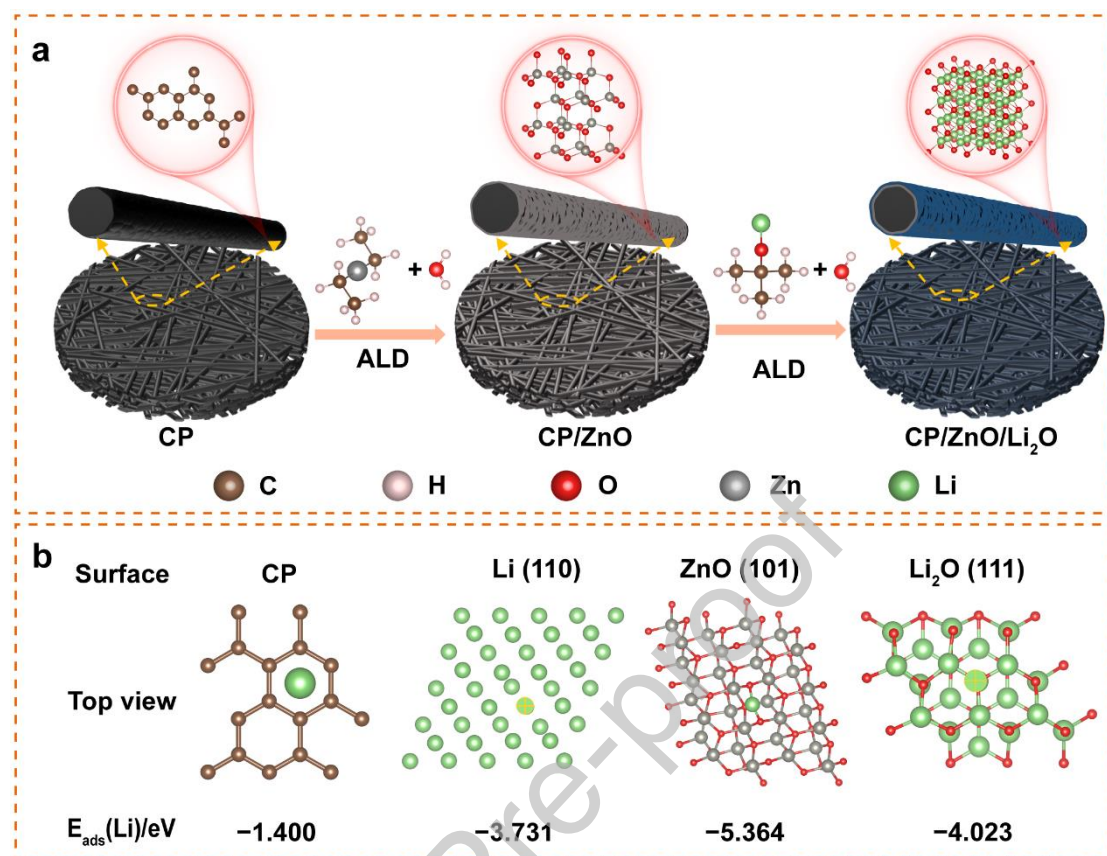


Fig. 1. (a) Schematic illustration of the preparation process of CP/ZnO/Li₂O current collector. (b) The Li adsorption configuration and the corresponding E_{ads} at the surface of CP, Li (110), ZnO (101), and Li₂O (111).

To systematically tailor the coating composition and structure, the thickness of the Li₂O layer was precisely adjusted by controlling the number of atomic layer deposition (ALD) cycles, yielding a series of 3D current collectors designated as CP/Z6, CP/Z6L0.5, CP/Z6L1, and CP/Z6L2. Scanning electron microscopy (SEM) revealed conformal coating coverage on carbon fibers, with surface morphology evolving from nanoparticles to nanosheets as the Li₂O thickness increased (Fig. 2a, d and Fig. S2). Digital images revealed a consistent color transition from gray-black to blue, confirming homogeneous surface modification. Energy-dispersive X-ray spectroscopy

(EDS) mapping confirmed uniform Zn and Li distribution across both the surface and cross-sectional regions of the carbon paper (Fig. 2b, c and Fig. S3). X-ray diffraction (XRD) patterns exhibited three distinct peaks in the 30–40° range (Fig. 2e), indexed to the hexagonal ZnO phase (JCPDS No. 36-1451), while the absence of Li₂O diffraction peaks suggests its amorphous or ultrathin character. X-ray photoelectron spectroscopy (XPS) analysis further confirmed the coating composition (Fig. 2g, h and Fig. S4). The Zn 2p spectrum showed peaks at 1044.08 eV and 1021.03 eV, corresponding to Zn 2p^{1/2} and Zn 2p^{3/2}, respectively, with a 23.1 eV energy separation characteristic of Zn²⁺. The O 1s spectrum was deconvoluted into two components at 530.0 eV and 531.1 eV, attributed to lattice oxygen (Zn–O/Li–O) and surface-adsorbed oxygen species, respectively. These results confirm the successful deposition of ZnO and Li₂O nanoparticles on carbon fibers via ALD. Four-point probe measurements showed a slight reduction in electronic conductivity from $1.3 \times 10^4 \text{ S m}^{-1}$ for pristine CP to $1.23 \times 10^4 \text{ S m}^{-1}$ after 200 ALD cycles, highlighting the electron-insulating nature of Li₂O (Fig. 2f). Ionic conductivity measurements revealed that the ZnO coating slightly decreased the ionic conductivity of CP, whereas the Li₂O coating significantly enhanced it (Fig. 2i and Fig. S5). These results demonstrate that Li₂O serves as an effective ionic conductor while maintaining electronic insulation, thereby contributing to the vertical conductivity gradient across the bilayer interface.

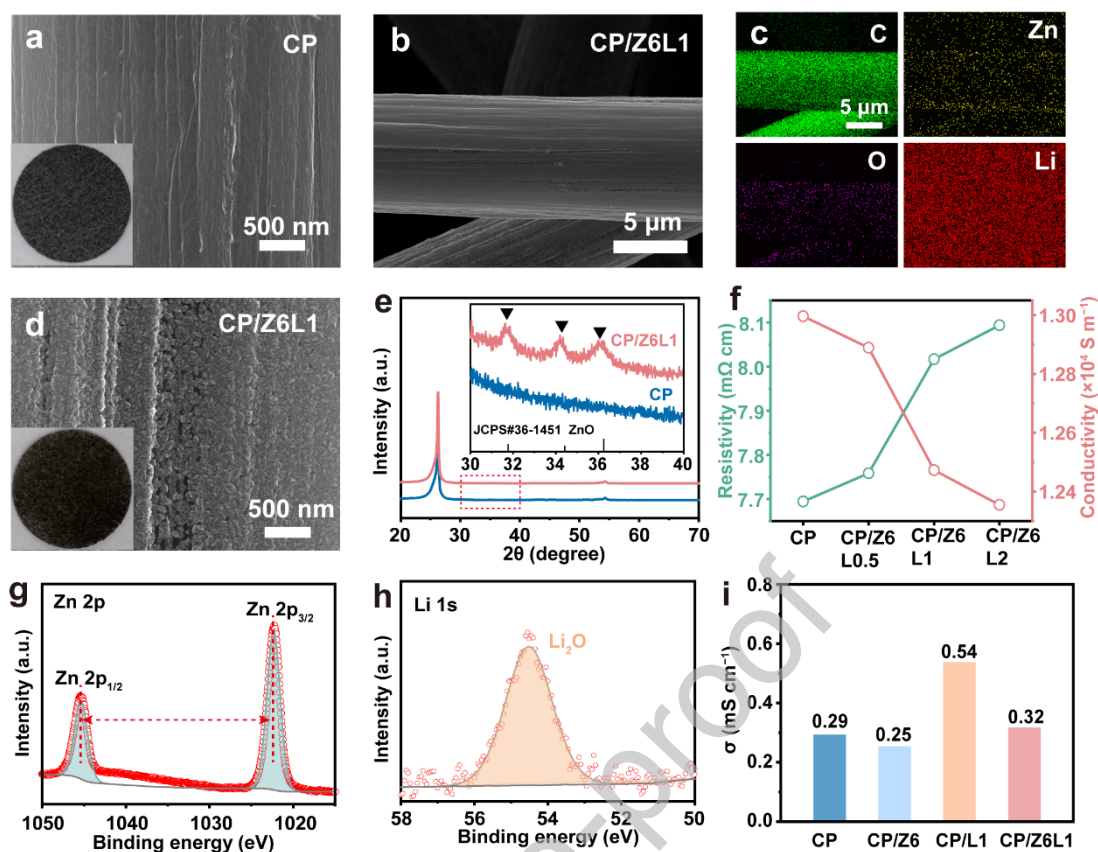


Fig. 2. SEM images and digital photos (inset) of (a) CP and (d) CP/Z6L1 samples. (b) SEM images, and (c) corresponding mapping images of CP/Z6L1 fiber. (e) XRD patterns of CP and CP/Z6L1 samples. (f) Electrical properties of CP/ZnO/Li₂O with various thicknesses deposition layers. XPS spectra of CP/Z6L1: (g) Zn 2p signal of ZnO, (h) Li 1s signal of Li₂O. (i) ion conductivity of various current collectors.

Phase evolution during Li deposition and stripping was investigated using in-situ XRD on Li||CP/Z6L1 half-cells (Fig. 3a). The diffraction patterns in the 23–27° region tracked the lithiation of carbon fibers, as evidenced by the reversible appearance and disappearance of LiC₁₂ and LiC₆ phases throughout the cycling process. During the initial discharge, ZnO-related peaks were observed between 30–40°, with intensities gradually decreasing and eventually disappearing. Simultaneously, a new diffraction

peak at 41.2° , corresponding to the Zn (101) plane, emerged and intensified, suggesting the transformation of ZnO to crystalline Zn via reaction with Li. As the process continued, the Zn (101) peak gradually diminished, while diffraction signals at 40.8° , 48.3° , and 74.4° , corresponding to the LiZn (110), (311), and (422) planes, respectively, became more pronounced, indicating the formation of Li–Zn alloys. These peaks exhibited excellent reversibility, implying that the in-situ generated LiZn alloy can act as both a Li reservoir and a nucleation site for subsequent plating cycles. Moreover, a peak corresponding to Li (110) emerged, accompanied by a decline in the intensities of Li (200) and Li (211), suggesting preferential horizontal Li migration along the (110) plane and promoting dendrite-free, planar deposition parallel to the substrate[32]. Although Li_2O forms during ZnO reduction, no Li_2O diffraction peaks were observed, indicating that the formed phase is amorphous or low crystallinity [24].

To predict the synergistic effect of the ZnO/ Li_2O bilayer, COMSOL simulations were performed to compare Li deposition behavior on CP/Z6 and CP/Z6L1 electrodes (Fig. 3b, c). In the CP/Z6 system, Li nucleation and growth were strongly localized, with lithiophilic sites driving protrusion formation and producing pronounced heterogeneity in the Li^+ concentration field, including depletion at protrusion tips and accumulation in recessed regions. Such gradients are expected to aggravate morphological instability at the electrode/electrolyte interface. In contrast, the ZnO/ Li_2O bilayer markedly homogenized the Li^+ flux: the Li_2O top layer facilitated uniform redistribution of Li^+ , while the ZnO-derived LiZn/ Li_2O domains at the bottom provided stable sites for ion–electron recombination. This vertically graded architecture transformed Li^+ transport

from a surface-diffusion dominated two-dimensional process to a bulk-diffusion dominated three-dimensional process, thereby promoting spatially uniform Li nucleation and the subsequent growth of a dense and smooth Li layer.

Li deposition morphology and thickness evolution under ZnO/Li₂O coatings were systematically studied by plating Li onto 3D frameworks at different areal capacities (Fig. 3d-g and Fig. S6). At 3 mAh cm⁻², bare Li formed coral-like dendrites, while Li on CP accumulated unevenly on the surface. In contrast, CP/Z6L1 enabled ultrathin, uniform Li layers with visible carbon fibers, indicating favorable nucleation. Even at 10 mAh cm⁻², CP/Z6L1 maintained dendrite-free, compact Li, minimizing electrolyte contact and parasitic reactions, whereas Li on CP formed loose dendrites and isolated islands, causing inactive Li accumulation. Cross-sectional SEM showed that depositing 10 mAh cm⁻² of Li on a planar Li substrate increased the thickness by ~140 μm, which is already comparable to the compressed CP thickness. Moreover, Li preferentially deposited on the top surface of lithiophobic CP, increasing the thickness from ~170 μm to ~230 μm. For lithiophilic CP/Z6L1, thickness remained almost unchanged, demonstrating that the ZnO/Li₂O coating guided bottom-up deposition and efficient utilization of the 3D scaffold, reducing volume fluctuations during cycling. Nucleation-size distributions were analyzed at a fixed current density of 0.5 mA cm⁻² with a deposition capacity of 5 mAh cm⁻² for Li, bare CP, and CP/Z6L1 electrodes (Fig. S8). The SEM images and statistical histograms show that Li nuclei formed on CP/Z6L1 are significantly larger than those on Li and bare CP. This is attributed to the higher interfacial resistance of the Li₂O layer, which reduces the density of nucleation sites

and promotes radial diffusion. The formation of larger nuclei decreases the effective surface area, thereby limiting electrolyte consumption and mitigating dead-Li accumulation.

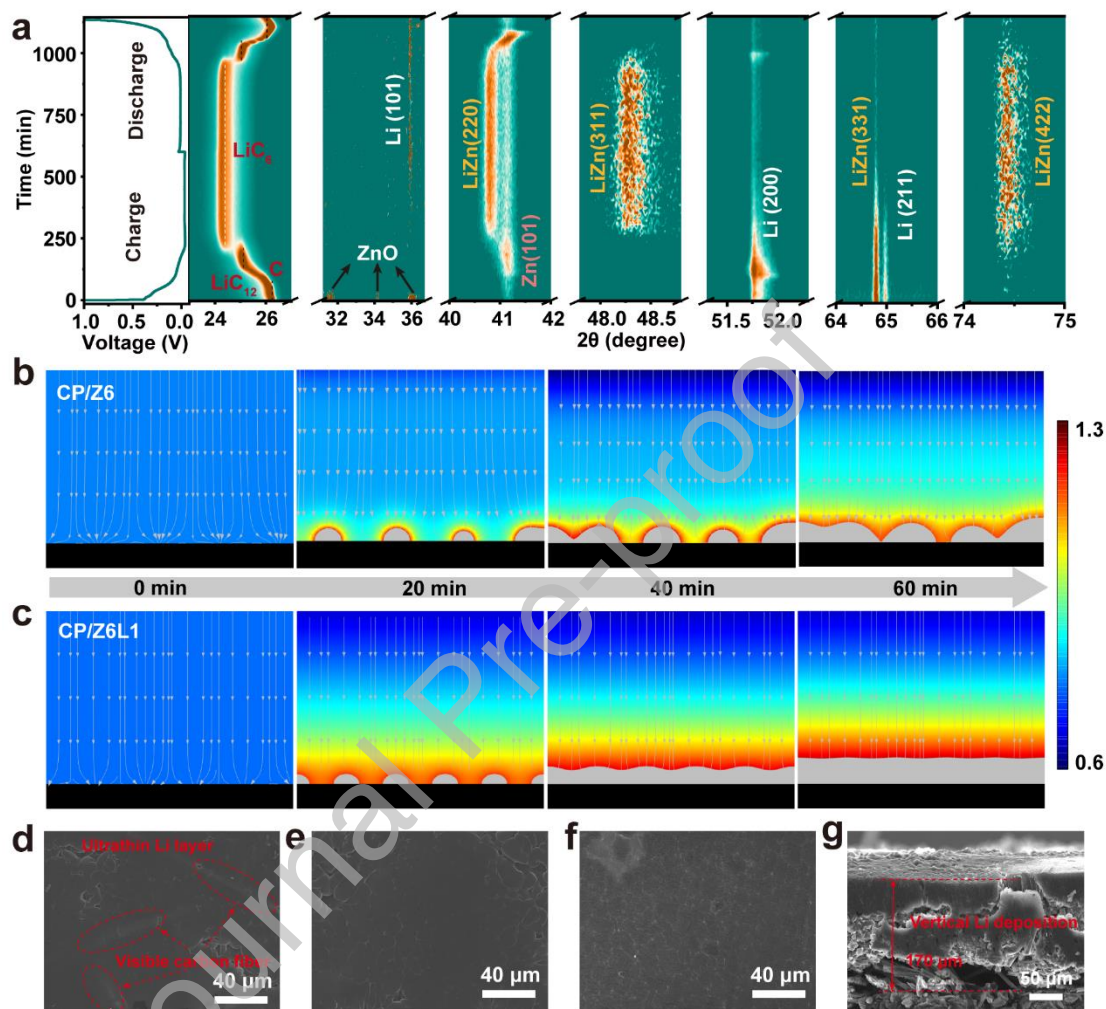


Fig. 3. (a) First discharge–charge curves of Li||CP/Z6L1 and corresponding in situ XRD pattern at a current density of 0.5 mA cm⁻². COMSOL simulation of the concentration field for (b) CP/Z6, and (c) CP/Z6L1 current collectors. SEM top-view images of CP/Z6L1 anodes after Li deposition with (d) 3 mAh cm⁻², (e) 5 mAh cm⁻², (f) 10 mAh cm⁻² at 0.5 mA cm⁻². SEM cross-section images of CP/Z6L1 anodes after Li deposition with (g) 10 mAh cm⁻² at 0.5 mA cm⁻².

The reversibility of Li plating/stripping in 3D current collectors was evaluated via

half-cell Coulombic efficiency (CE) and average CE (ACE), with the latter calculated using the Aurbach method to optimize Li_2O coating thickness. CP/Z6L1 exhibited the highest ACE of 98.0% and ultralow nucleation overpotential of 14 mV, significantly outperforming bare CP (45 mV), indicating enhanced lithiophilicity and uniform Li nucleation (Fig. S9, 10). Li@CP/Z6L1 symmetric cells maintained low overpotentials (~ 10 mV) over extended cycling exceeding 2400 h at $0.5 \text{ mA cm}^{-2}/0.5 \text{ mAh cm}^{-2}$, whereas Li and Li@CP cells failed after 310 h and 1000 h due to severe voltage fluctuations (Fig. 4a). The superior rate performance of Li@CP/Z6L1 was further demonstrated under $0.5\text{--}8 \text{ mA cm}^{-2}$, with overpotentials increasing moderately from ~ 10 mV to 40 mV, compared to 75 mV for Li@CP , reflecting a highly ion-conductive SEI and efficient Li^+ transport (Fig. 4b, c). Electrochemical impedance spectroscopy confirmed consistently lower SEI (R_{SEI}) and charge-transfer (R_{ct}) resistances for Li@CP/Z6L1 , indicating the formation of a thinner, more stable interphase during cycling (Fig. S11).

Low-temperature electrochemical performance of Li deposition was evaluated using half cells and symmetric cells. At -20°C , CP/Z6L1 electrodes exhibited higher CE and superior cycling stability compared to bare CP. Specifically, CP electrodes failed after ~ 90 cycles, whereas CP/Z6L1 maintained stable cycling for up to 165 cycles (Fig. 4d). The Li@CP/Z6L1 electrodes also showed markedly lower overpotentials than Li and Li@CP electrodes, indicating enhanced interfacial kinetics (Fig. 4e). Control experiments with single-layer ZnO (CP/Z6) and Li_2O (CP/L1) coatings revealed that the bilayer CP/Z6L1 consistently achieved lower polarization and more stable cycling

under both room temperature and $-20\text{ }^{\circ}\text{C}$ conditions (Fig. 4f and Fig. S12), demonstrating the synergistic effect of the ZnO/Li₂O bilayer in promoting uniform Li deposition and interfacial stability. These electrochemical results are consistent with the predictions of COMSOL simulations.

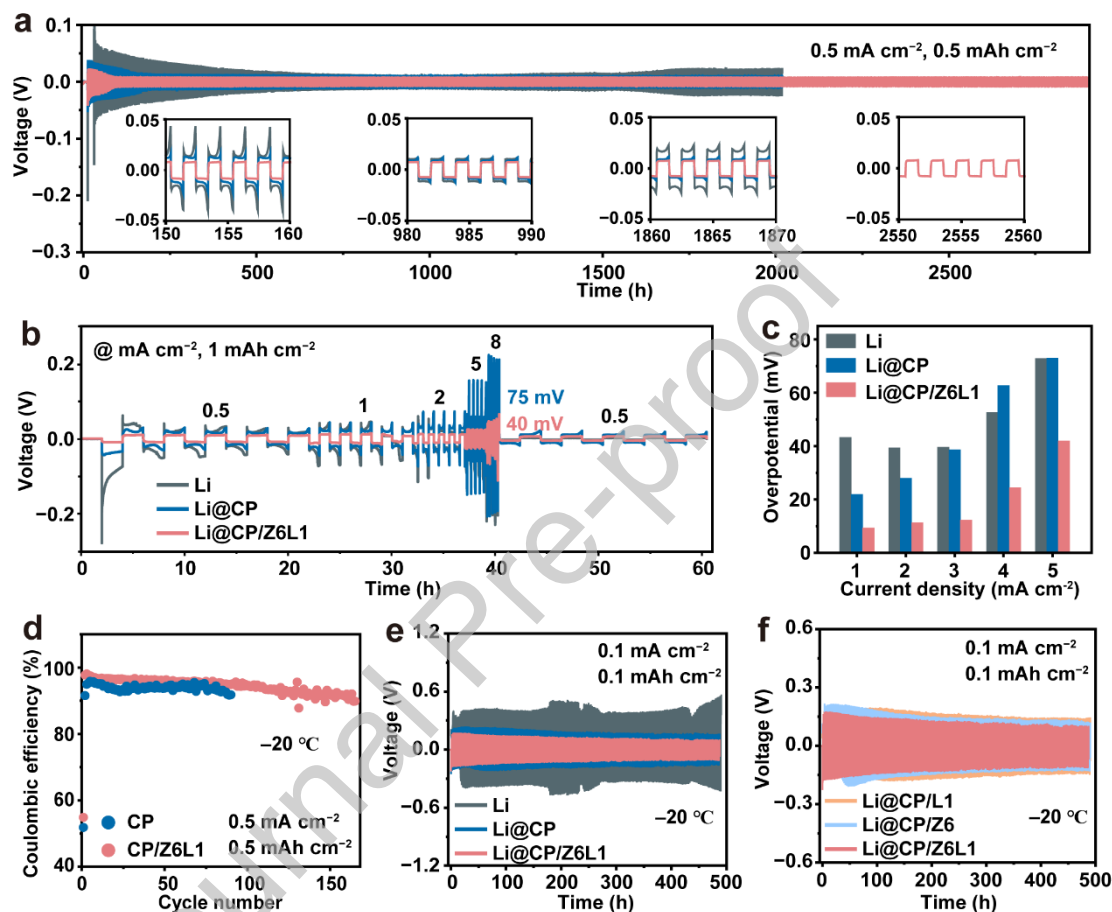


Fig. 4. (a) The voltage profiles of Li, Li@CP, and Li@CP/Z6L1 symmetrical cells at a current density and areal capacity of 0.5 mA cm^{-2} and 0.5 mAh cm^{-2} . The insets show the high-resolution voltage profiles at specific times. (b) Rate performances at the current densities from 0.5 to 8 mA cm^{-2} with 1 mAh cm^{-2} . (c) Variation of the Li plating/stripping overpotential on various current densities using Li, Li@CP, and Li@CP/Z6L1. (d) CEs of Li plating/stripping using bare CP, and CP/Z6L1 at a current density and areal capacity of 0.5 mA cm^{-2} and 0.5 mAh cm^{-2} at $-20\text{ }^{\circ}\text{C}$. (e, f) The voltage profiles

of symmetrical cells at a current density and areal capacity of 0.1 mA cm^{-2} and 0.1 mAh cm^{-2} at -20°C .

To investigate the evolution of SEI composition, XPS depth profiling was performed on cycled electrodes (Fig. S14-S17). As shown in Fig. 5a, d, the C 1s spectra consisted of C–C (284.8 eV), C–O (286.4 eV), C=O (288.5 eV), Li_2CO_3 (290.2 eV), C–F₂ (290.9 eV), and C–F (292.9 eV), originating from electrolyte decomposition. After 60 s of Ar^+ sputtering, carbonaceous species were significantly reduced on the CP/Z6L1 surface, with residual CO_3^{2-} peaks remaining, whereas stronger organic signals persisted on CP/Z6 and bare Li. High-resolution F 1s spectra (Fig. 5b, e) revealed the coexistence of organic C–F (688.8 eV) and inorganic LiF (685.2 eV). A pronounced LiF signal was observed on CP/Z6L1, which further increased after sputtering, while C–F components disappeared. In contrast, CP/Z6 contained less LiF and retained noticeable C–F signals. Quantitative analysis confirmed that the CP/Z6L1 electrode contained the highest fraction of inorganic species, dominated by LiF and Li_2CO_3 , whereas CP/Z6 and bare Li exhibited higher proportions of organic species (Fig. 5c, f). These results indicate that the bilayer ZnO/ Li_2O interface promotes more complete decomposition of TFSI[−] and facilitates the formation of a LiF-rich SEI, consistent with improved interfacial stability and Li^+ transport. In addition, SEM observations after 30 plating/stripping cycles at 5 mAh cm^{-2} revealed that bare CP was fully covered by a dense Li layer with filament-like deposits and thick residual Li after stripping. In contrast, CP/Z6L1 exhibited uniform, dendrite-free deposition, with Li preferentially filling the internal scaffold and leaving minimal residual Li after stripping, thus exposing the 3D

framework (Fig. S18).

In ether-based electrolytes, the low LUMO energy of LiTFSI (-1.61 eV) favors its preferential reduction and contributes to the formation of LiF-rich SEI. To further elucidate the interfacial mechanism, DFT calculations were carried out[35]. The adsorption energy of TFSI⁻ on bare Li (-2.8 eV) was significantly less negative than those on Li₂O (111) (-3.565 eV) and LiZn (111) (-5.185 eV), indicating stronger binding of TFSI⁻ at the coated interfaces (Fig. 5g). Differential charge density analyses revealed charge transfers of 1.04 electrons from Li₂O to TFSI⁻ (Fig. 5h), consistent with electrostatic potential mapping that identified terminal O–Li groups in Li₂O as electron-rich sites with strong nucleophilicity (Fig. 5i). These regions readily interact with the positively polarized S=O and C–F bonds in TFSI⁻, thereby promoting its stepwise decomposition into LiF [3, 36]. In this dual-layer artificial interface, the Li₂O top layer acts as an ionic conductor and electronic insulator, blocking electron penetration while permitting Li⁺ transport, whereas the underlying LiZn layer provides mixed ionic/electronic conductivity that facilitates Li⁺/electron recombination (Fig. 5j). This vertical conductivity gradient not only guides bottom-up Li deposition and enables uniform, dendrite-free growth, but also promotes selective decomposition of TFSI⁻ into LiF, as further supported by XPS analysis.

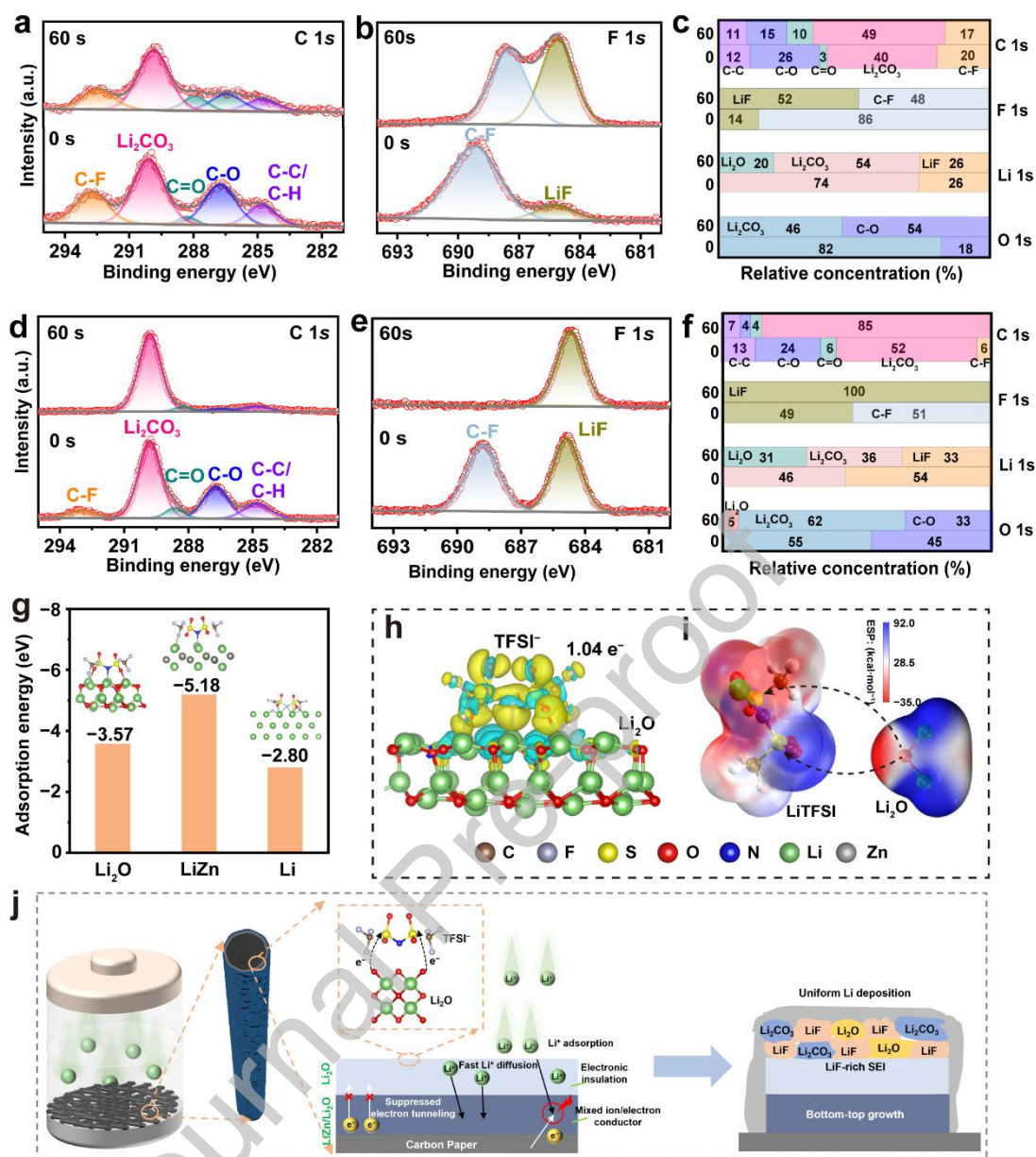


Fig. 5. (a) C1s, and (b) F1s high-resolution XPS depth spectra of Li@CP/Z6 symmetric cells after the 30th cycle. (d) C1s, and (e) F1s high-resolution XPS depth spectra of Li@CP/Z6L1 symmetric cells after the 30th cycle. Schematic representation of the proportions of the individual elements in the SEI and the SEI components of (c) Li@CP/Z6, and (f) Li@CP/Z6L1. (g) Calculated adsorption energies of TFSI⁻ on Li₂O, LiZn and Li substrates. (h) Differential charge density images of Li₂O with TFSI⁻. Charge accumulation: yellow areas, and charge depletion: cyan areas. (i) Electrostatic

potential distribution of TFSI^- and Li_2O at the isosurface of electron density. (j)

Schematic illustration of the Li deposition behavior regulated by the CP/Z6L1 artificial interfacial layer.

In-situ electrochemical impedance spectroscopy (EIS), combined with distribution of relaxation times (DRT) analysis, was utilized to track the dynamic evolution of Li@CP/Z6L1||LCO full cells during the initial charge/discharge cycles (Fig. 6a, and d). DRT deconvolution resolved five characteristic relaxation times (τ_1 , τ_2 , τ_3 , τ_4 , and τ_5) corresponding to ohmic resistance (R_b), ion transport through anode and cathode interfaces resistances (R_{SEI} and R_{CEI}), anode charge transfer resistance ($R_{\text{ct-A}}$), cathode charge transfer resistance ($R_{\text{ct-C}}$), and diffusion resistance (R_D). The impedance values presented in Fig. 6c and 6f were obtained by integrating the areas under the DRT curves shown in Fig. 6b and 6e. Compared to the Li||LCO, both Li@CP||LCO and Li@CP/Z6L1||LCO exhibited significantly lower impedance values across all frequency regions. Impedance integration analysis (Fig. S19 and S20) revealed negligible changes in R_b and R_{SEI} with varying states of charge (SOC), confirming the exceptional stability of the SEI layer derived from the Li@CP/Z6L1 interface. The fast ion-conductive SEI facilitates uniform Li nucleation across the entire 3D scaffold surface. The charge transfer process at the anode interface is the rate-limiting step of the entire electrochemical reaction. Notably, as Li deposition proceeded during charging, the intensity of the $R_{\text{ct-A}}$ decreased, and two distinct semicircles progressively merged into a single peak. This indicates the completion of the electrochemical reaction and the successful construction of the 3D Li – carbon

composite anode. During discharge, the intensity and evolution trend of the Rct-A in the Li@CP/Z6L1 exhibit a reversible behavior compared to that during charging, indicating excellent reversibility of the Li plating/stripping process at the Li@CP/Z6L1 anode. In contrast, a sharp impedance increase was readily apparent in the bare Li cell, which can be attributed to the accumulation of electrically isolated dead Li, ultimately resulting in interfacial contact loss. These findings highlight the essential role of the lithiophilic artificial interface in enhancing charge transfer kinetics and modulating SEI behavior.

The reduced activation energy for charge transfer further corroborates this conclusion, reflecting faster reaction kinetics at the electrode (Fig. 6g and Fig. S21). Specifically, the activation energy for Li^+ transport through the SEI on the Li@CP/Z6L1 electrode ($6.32 \text{ kJ}\cdot\text{mol}^{-1}$) is substantially lower than that of bare Li ($12.21 \text{ kJ}\cdot\text{mol}^{-1}$) and Li@CP ($14.50 \text{ kJ}\cdot\text{mol}^{-1}$) (Fig. 6h), aligning with the presence of a LiF-rich, ionically conductive SEI layer observed in the XPS analysis. Furthermore, Tafel analysis demonstrates a significantly enhanced exchange current density for Li@CP/Z6L1 (0.7 mA cm^{-2}) compared to CP (0.44 mA cm^{-2}), reinforcing the improved interfacial dynamics and superior Li deposition behavior (Fig. 6i).

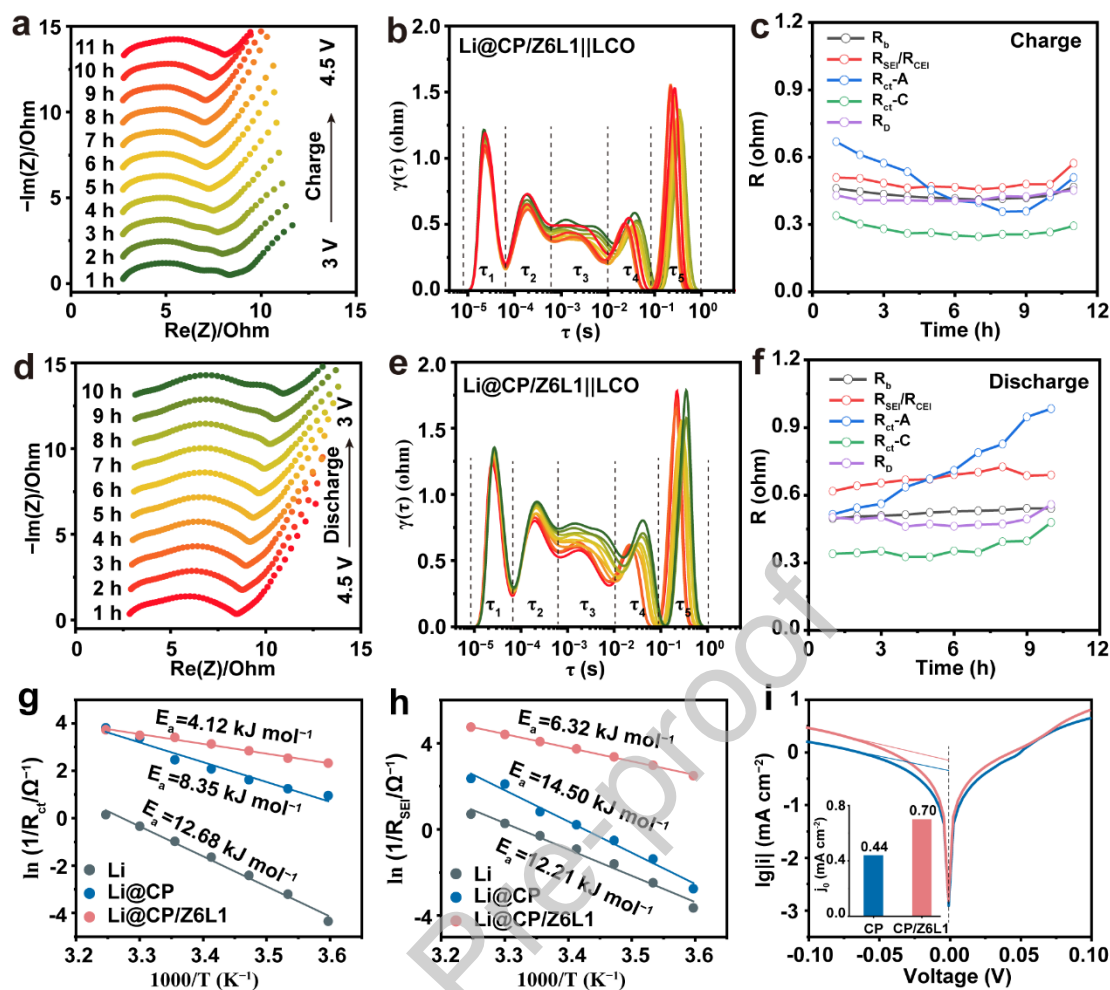


Fig. 6 (a, d) Nyquist plots of Li@CP/Z6L1||LCO cells at 0.1 C during the 1st cycle of charge and discharge. (b, e) Corresponding DRT calculated from EIS. (c, f) Time evolution of Li@CP/Z6L1||LCO resistance obtained by fitting every individual spectrum of DRT. The activation energies derived by Arrhenius fitting for (g) R_{ct} and (h) R_{SEI} with Li, Li@CP, and Li@CP/Z6L1 anode. (i) Tafel curves of CP, and CP/Z6L1 half cells and the calculated exchange current density (j_0).

The energy storage capability of the Li@CP/Z6L1 anode was further evaluated in full-cell configurations with both medium-loading and high-loading LCO cathodes. Cyclic voltammetry (CV) scans during the first 10 cycles revealed that the Li@CP/Z6L1||LCO full cell exhibited a smaller redox potential difference (117 mV)

and a lower charging voltage (3.98 V), indicating enhanced Li^+ transport kinetics and more reversible redox reactions (Fig. 7a and Fig. S22). When paired with a medium-loading LCO cathode (8.23 mg cm^{-2}), the Li@CP/Z6L1 anode demonstrated superior cycling stability, retaining 70.1% of its initial capacity after 400 cycles, compared to only 34% for the Li||LCO (Fig. 7b and Fig. S23). The Li@CP/Z6L1 anode delivered 178.3, 159.5, and 136.7 mAh g^{-1} at 1, 3, and 5 C, respectively, whereas the Li||LCO cell suffered severe capacity decay (83.8 mAh g^{-1} at 5 C) (Fig. 7c and Fig. S24). The long-term cycling performance was further examined at -20°C . The Li@CP/Z6L1||LCO full cell delivered an initial discharge capacity of 178.3 mAh g^{-1} and retained 83.1% of its capacity after 200 cycles ($\sim 4000 \text{ h}$), significantly outperforming the bare Li cell (Fig. 7d, e and Fig. S25). The dQ/dV analysis (Fig. S25c) revealed a pair of redox peaks in the 4.1–4.3 V region, consistent with the oscillations observed in GCD curves, which originate from the intrinsic order–disorder phase transition of LiCoO_2 near $\text{Li}_{0.5}\text{CoO}_2$ ($x \approx 0.5$)[37–39]. These features confirm that the voltage fluctuations are a manifestation of structural transitions intrinsic to LCO rather than instability of the bilayer interface. Against state-of-the-art low-temperature Li||LCO full cells, the CP/Z6L1 anode delivers superior discharge capacity and capacity retention, highlighting its competitiveness under harsh conditions (Fig. 7f and Table 4). Furthermore, a Li@CP/Z6L1||LFP pouch cell successfully powered an LED even at -20°C (Fig. 7g), confirming the practical applicability of this design in low-temperature environments.

To probe more practical conditions, full cells with higher-loading LCO cathodes

(10.89 mg cm⁻², 1.94 mAh cm⁻²) were cycled at 2 C (Fig. 7h and Fig. S26). Under these stringent conditions, bare Li and bare CP electrodes exhibited rapid capacity fading, while the CP/Z6 electrode showed moderate improvement. In contrast, CP/Z6L1 retained 111.6 mAh g⁻¹ after 75 cycles (85.6% retention), underscoring its superior robustness at high loading and current density. Compared with reported Li||LCO full cells, the CP/Z6L1 electrode operates at higher areal capacities and current densities with markedly improved capacity retention, demonstrating its practical advantage (Fig. 7i and Table 5). The versatility of the bilayer coating was further validated in Li@CP/Z6L1||LFP full cells, which delivered an initial discharge capacity of 137.5 mAh g⁻¹ at 1 C and maintained 96.4% capacity after 500 cycles, with an average CE of 99.94% (Fig. S27). Benchmarking against recently reported Li||LFP full cells (Fig. S28 and Table 6) demonstrates that the bilayer-coated electrodes exhibit superior cycle life and capacity retention. In addition, the superiority of the Li@CP/Z6L1||LFP full cell was also verified in carbonate-based electrolytes (Fig. S29). Overall, these results highlight the critical role of the ZnO/Li₂O bilayer in stabilizing interfacial reactions and guiding bottom-up Li deposition, enabling high-rate capability, excellent durability under high-loading conditions, and outstanding low-temperature performance that surpasses state-of-the-art reports.

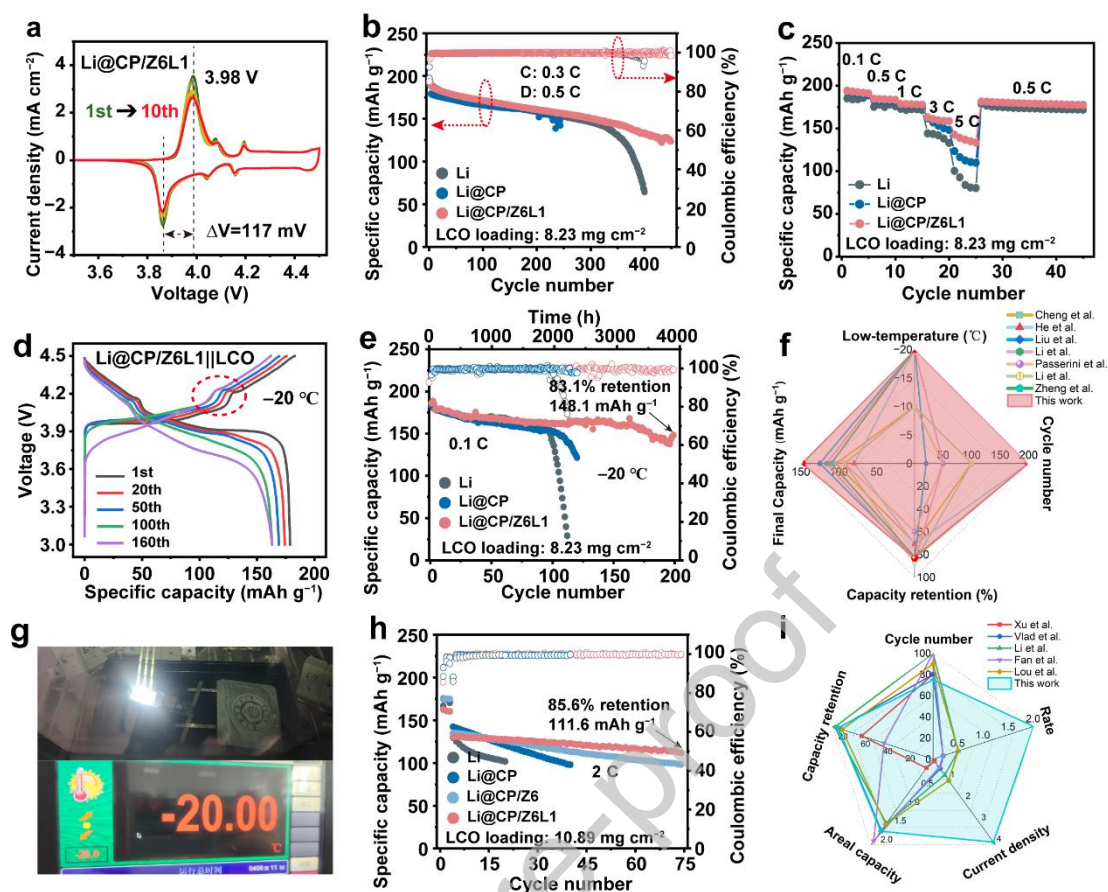


Fig. 7. (a) CV curves of the first 10 cycles for Li@CP/Z6L1||LCO full cells at a scanning rate of 0.1 mV s⁻¹. (b) Cycling performance and (c) Rate performance of Li||LCO, Li@CP||LCO, and Li@CP/Z6L1||LCO full cells at 25 °C. (d) Galvanostatic charge-discharge profiles of Li@CP/Z6L1||LCO cells at -20 °C. (e) Cycling performance of Li||LCO, Li@CP||LCO, and Li@CP/Z6L1||LCO full cells at -20 °C. (f) Comparison of electrochemical performance of Li@CP/Z6L1||LCO full cells at -20 °C with related work. (g) Photograph of the pouch cell lighting up an LED at -20 °C. (h) Cycling stability of full cells with high-loading LCO cathodes at 2C. (i) Comparison of electrochemical performance of Li@CP/Z6L1||LCO full cells at -20 °C with with previously reported work.

3. Conclusions

In summary, ZnO/Li₂O bilayer artificial interface on 3D carbon paper forms a vertically graded artificial interface that synergistically regulates Li nucleation and growth. The underlying ZnO-derived LiZn/Li₂O domains provide mixed ionic-electronic conductivity and stable nucleation sites, promoting directional Li deposition, while the high-resistance Li₂O top layer reduces the nucleation site density and facilitates radial Li diffusion. The formation of larger Li nuclei decreases the effective surface area, mitigating electrolyte consumption and dead-Li accumulation. The vertical conductivity gradient of the bilayer interface acts synergistically to guide uniform bottom-up Li deposition, resulting in a dense, planar Li layer. Additionally, the Li₂O layer induces complete decomposition of TFSI⁻, forming a LiF-rich SEI that stabilizes the interface and enhances Li⁺ transport. As a result, the Li@CP/Z6L1 symmetric cell achieves ultrastable cycling over 2900 h with minimal polarization. Full cells exhibit excellent performance under low-temperature and long-term cycling, with the Li@CP/Z6L1||LCO cell retaining 83.1% capacity at -20 °C after 200 cycles, and the Li@CP/Z6L1||LFP cell maintaining 96.4% capacity over 500 cycles. This scalable bilayer design provides a practical strategy for constructing functional artificial interphases and offers guidance for the development of high-performance lithium metal anodes.

Experimental section

Experimental details can be found in the Supporting Information.

Conflicts of interest

The authors declare no competing conflict of interest.

Acknowledgements

This work was supported by the National Key Research and Development Program of China (2022YFE0109400), Leading Edge Technology of Jiangsu Province (BK20220009, BK20232022), Priority Academic Program Development of Jiangsu Higher Education Institutions (PAPD), and Center for Microscopy and Analysis at Nanjing University of Aeronautics and Astronautics.

Appendix A. Supplementary material

Supplementary material to this article can be found online at...

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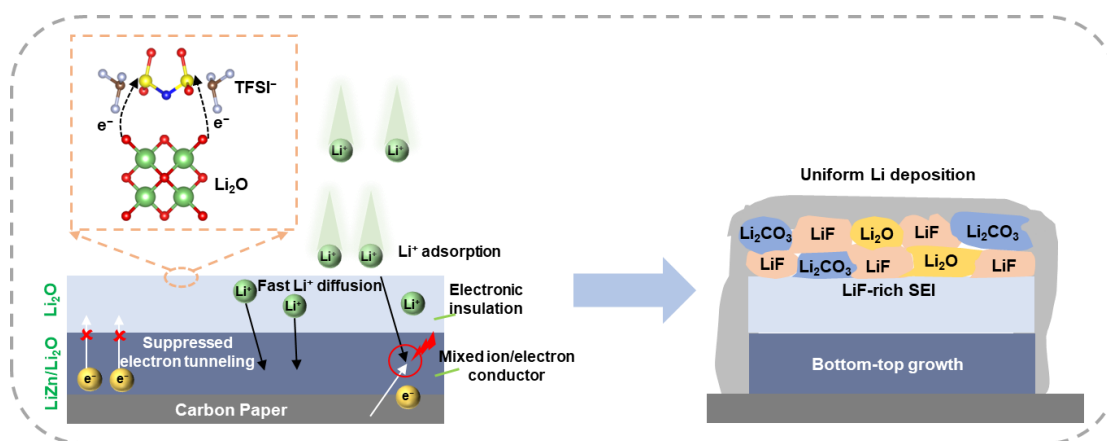
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Graphical abstract



Declaration of interests

☒ The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.